MIL-S-7877B
20 AUGUST 1956

Superseding
MIL-S-7877A
9 February 1953

MILITARY SPECIFICATION

SEATS; PASSENGER, CABIN FOLDING

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope.- This specification covers the general requirements for a standard 20g aircraft folding-type, passenger seat which may be installed in either a forward or aft facing position in all types of passenger and cargo aircraft.

1.2 Classification.- Passenger seats shall be one of the following three types, as specified (see 6.2):

Type MP-1 - Single
Type MP-2 - Double
Type MP-3 - Triple

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standard, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

NN-P-515	Plywood, Container Grade
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Plating (Electrodeposited)
TT-C-595	Colors; (For) Ready-Mixed Paints
CCC-A-700	Artificial Leather, Upholstery (Synthetic Resin Coated, Cloth)
PPP-B-601	Boxes, Wood, Cleated-Plywood

Military

JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed
JAN-P-108	Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior
MIL-P-116	Preservation, Methods of
JAN-P-125	Packaging and Packing for Overseas Shipment - Barrier-Materials, Waterproof, Flexible

FSC 1680

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JAN-P-127	Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water Resistant
MIL-A-140	Adhesive, Water-Resistant, Waterproof Barrier-Material
MIL-R-5001	Rubber; Latex Foam Sponge
MIL-D-5028	Drawings and Data Lists: Preparation of Manufacturers' (for Production Aircraft, Guided Missiles, Engines, Accessories, and Other Auxiliary Equipment)
MIL-B-8607	Belt, Safety, Passenger Type
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-L-10547	Liners, Case, Waterproof

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Air Force-Navy Aeronautical Standard Drawings

AN7516	Fitting - Cargo Airplane Tie Down
AND10398	Metals - Definition of Dissimilar

PUBLICATIONS

Air Force-Navy Aeronautical Bulletins

No. 143	Specifications and Standards; Use of
No. 157	Colors; List of Standard Aircraft Camouflage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids, shall apply.

Society of Automotive Engineers

AMS3676	Insulation, Sound and Thermal, Resin-Bonded Glass Fiber, Medium Filament

(Application for copies should be addressed to the Society of Automotive Engineers, Inc, 29 West 39th Street, New York 18, New York.)

3. REQUIREMENTS

3.1 Qualification.- The aircraft folding-type, passenger seats furnished under this specification shall be a product which has been tested and has passed the Qualification tests specified herein.

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3.2 Materials.- The basic structure of the seat shall be fabricated of metal or an acceptable substitute. Wood or other materials which are subject to attack by mildew or fungus shall not be used in the basic structure.

3.2.1 Protective treatment.- When materials are used in the construction of the seat that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of dissimilar metals, as defined in Drawing AND10398, in intimate metal-to-metal contact shall be avoided wherever practicable.

3.2.2 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.2.2.1 Standard parts.- Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design and construction.-

3.3.1 Design.- The seats shall be designed for maximum comfort, minimum weight, minimum over-all dimensions in both normal and folded position, interchangeability, and shall permit quick installation and removal.

3.3.2 Dimensions.- The outline dimensions of the seats shall be as specified in figures 1 and 3.

3.3.3 General structure.- Thermoseal straps, or equivalent, may be used on the back and bottom assemblies to form a resilient structure for the back and bottom cushion assemblies. The seat and back shall have the maximum developed posture contours, with the use of formed cushions or padding, if desirable.

3.3.4 Construction methods.- Riveting or welding may be used for assembly of component parts fabricated of metals which are suitable for this type of construction. Fittings and joints which will require disassembly of the component parts of the seat shall be bolted.

3.3.5 Projections.- The inside surface of the seat shall be free from projections to prevent snagging or damage to clothing or seat cushions and upholstery. The exterior surfaces of the seat shall be free from sharp edges or any projections which might injure the occupant or damage clothing during ingress or egress from seats. Individual covers shall be provided at front and rear of seat frame to cover leg attachment fitting when leg is folded.

3.3.6 Strength requirements.- The seats shall be used either in a forward or aft facing position and shall have sufficient ultimate structural strength to carry the loads as specified in figures 2, 4, and 5 without failure or excessive deflection.

3.3.6.1 Seats shall take specified loads with legs positioned or folded within limits specified in figures 1 through 5.

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3.3.6.2 Seat back.- Each seat back load (as specified in figures 2, 4, and 5) shall be uniformly distributed over each seat back and acting rearward in a horizontal plane, under three separate conditions, as follows: When leg arrangements are symmetrical with respect to the applied load, either (b) or (c) may be used.

(a) Directly aft.
(b) 20 Degrees to the right.
(c) 20 Degrees to the left.

3.3.7 Safety belts.- Provisions shall be made for the installation of a suitable 20g passenger-type lap belt. The belt shall be in accordance with Specification MIL-B-8607.

3.3.8 Floor attachment.- Each leg of the seat assembly shall be provided with a quick-disconnect fitting assembly which shall not exceed 1.25 inches in diameter for the

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5.3 Packing.- Each shipping container, insofar as possible, shall contain the the identical number of seats, shall be of a uniform size, and shall be designed to enclose the contents in a snug, tight-fitting manner. The gross weight of the shipping container, when packed for shipment, shall not exceed 200 pounds, whenever practicable. Unless otherwise specified, the interior packages shall be packed in cleated-plywood boxes, conforming to Specification PPP-B-601, except that the plywood shall be type II, class 2, conforming to Specification NN-P-515, or in nailed wood boxes in accordance with Specification JAN-P-106. Shipping containers conforming to JAN-P-106 shall have case liners conforming to Specification MIL-L-10547 and appendix thereto. Box closures shall be as specified in the appendix to the applicable box specification.

5.4 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The identification shall be composed of the following information listed in the order shown:

Stock No. or other identification number as specified
in the purchase document
SEATS; PASSENGER, CABIN FOLDING
Type (MP-1, MP-2, MP-3, as applicable)

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6.3.2 Qualified Products List.- For procurement purposes, the provisions of this specification will not become effective until publication of the Qualified Products List, to be numbered QPL-7877B-1.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army - Transportation Corps
Navy - Bureau of Aeronautics
Air Force

Preparing activity:
Navy: Bureau of Aeronautics

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 1.9-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE C-0870

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PLATE NO. 13419 (BACK)

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